

EDK II PEI dispatcher hang in Dasharo payload

The Coreboot build logs show that the EDK II PEI Core began dispatching modules (PEIMs) but never reached DXE; it printed “End of peim entry-1” and “End of DelayedDispatch” and then stopped. In other words, it finished the first PEIM and entered the **Delayed Dispatch** loop, but no further progress was made. In EDK II, the PEI dispatcher will wait for any registered delayed-dispatch entries to complete (or time out) before continuing [10†L2629-L2640]. If a PEIM has registered a delayed callback that never finishes or is never triggered, the dispatcher can appear to “hang” indefinitely (up to a built-in timeout). This matches the observed freeze: the firmware is sitting in the DelayedDispatch loop, waiting on a task that never completes.

Likely culprit: stalled delayed PEIM

By inspecting the payload’s FV ([DasharoPayloadPkg.fdf](#)), we see 15 PEI modules in order (SecCore, PeiMain, Pcd, SmmStorePei, FaultTolerantWritePei, BtSupportPei, ReportStatusCodeRouterPei, StatusCodeHandlerPei, DxeIpl, FirmwarePerformanceDataTablePei, Tcg2ConfigPei, TcgPei, Tcg2Pei, OpalPasswordPei, HddPasswordPei). The PEIMs run in the FV order by default [13†L105-L112]. The debug “End of peim entry-1” implies only the first PEIM executed. Immediately after, “End of DelayedDispatch” is printed, meaning the core entered the delayed-dispatch phase and returned without dispatching any more modules. (If the loop had dispatched another module, we would have seen “End of peim entry-2”, etc.)

Which module could have registered a delayed callback? Common suspects are the TPM (TCG) or SMM store modules. For example, **TCG2** initialization often uses a delayed call to poll the TPM or establish a TPM event, and **SmmStorePei** may schedule background tasks. If such a module set up a delayed entry and never signals it (perhaps due to missing hardware or a configuration issue), the core will wait until the retry timeout (which by default could be large). In the TianoCore code (Dispatcher.c), the delayed loop runs “while (DelayedDispatchTable->Count > 0) ...” and only exits on timeout [10†L2629-L2640]. This effectively stalls further PEI progress.

The log fragment suggests exactly this: after the first entry, the core printed status codes (the lines “n-0,1,0,0” and “L-1,1,B,0,1,0,0” are progress-code output), then nothing. In TianoCore’s dispatcher, if delayed entries remain, the loop won’t return until they finish or timeout [10†L2629-L2640]. In other words, the firmware *is* running but waiting on a PEI callback that never fires.

Conclusion

In summary, the Dasharo payload appears to be hung in the PEI phase, specifically in the delayed-dispatch loop, after completing only one PEIM. This is why “EDK II is not executed successfully” – it’s not advancing to DXE because a PEIM has effectively frozen the dispatcher. The root cause is likely a PEIM (e.g. a TPM2/TCG module or the SMM store module) that registered a delayed dispatch callback and never completed it. As the EDK II code documents, the dispatcher will wait indefinitely for delayed entries [10†L2629-L2640].

Sources: TianoCore EDK II PEI Core Dispatcher implementation, which shows how delayed dispatch entries are handled and can block progress 【10†L2629-L2640】 ; TianoCore documentation on PEI module ordering (PEIMs run in FDF/DSC order) 【13†L105-L112】 . These explain why the payload log stops at “End of DelayedDispatch”.
